Newman Fabian Monrose

Julian T. Hightower Chair
School of Electrical and Computer Engineering

[I] Degrees

Ph. D., Computer Science, — *Courant Institute of Mathematical Sciences*
New York University, New York, USA — May, 1999
Advisor: Prof. Zvi Kedem

M. Sc., Computer Science, — *Courant Institute of Mathematical Sciences*
New York University, New York, USA — May, 1996

B. Sc., Computer Science, — *Barry University*
Miami, Florida, USA — May, 1993

[II] History

Julian T. Hightower Chaired Professor, — *Georgia Tech*
College of Engineering, — August 2022 — present

Kenan Distinguished Professor, — *University of North Carolina, Chapel Hill*
Computer Science Department, — July 2017 — July 2022

Co-Founder, — *Zeropoint Dynamics*
Role: Chief Scientist, — March 2015 — 2022

Dir., Computer & Information Security — *Renaissance Computing Institute*
Joint appointment with UNC-CH Comp. Sci. — January 2014 — 2018

Professor, — *University of North Carolina, Chapel Hill*
Computer Science Department, — January 2013 — present

Associate Professor, — *University of North Carolina, Chapel Hill*
Computer Science Department, — July 2008 — December, 2012

Associate Professor, — *Johns Hopkins University*
Computer Science Department, — July 2007 —June 2008

Assistant Professor, — *Johns Hopkins University*
Computer Science Department, — November 2002 — June 2007

Research Scientist, — *Bell Laboratories, Lucent Technologies*
Secure Systems Research, — July 1999 — October 2002

[III] Honors

- Distinguished Reviewer, *ACM CCS Symposium*, — Fall, 2025
- CIOS Honor Roll, *Center for Teaching and Learning*, — Fall, 2023
- CSSA Graduate Teaching Award, *CS Department*, — May, 2021
- Undergraduate Students Teaching Award, *CS Department*, — May, 2015
- Best Student Paper Award, *IEEE Symposium on Security & Privacy* — May, 2013

- Outstanding Research in *Privacy Enhancing Technologies* (PET), July, 2012
- AT&T Best Applied Security Paper Award, *NYU-Poly CSAW*, Nov, 2011
- Best Paper Award, *IEEE Symposium on Security & Privacy*, May, 2011
- Best Paper Award, *Intl. Conf. on Internet Monitoring and Protection*, April, 2011
- Faculty Research Award, *Google* May, 2011

[IV] Scholarship
(*last 15 yrs only***)**

Submitted Articles

- *Beyond Reach: Measuring Threat Intelligence Gaps on Shared Domain Hosting Platforms**. **Tillson Galloway**, Evangelos Froudakis and Fabian Monrose. Submitted to ACM Web Conference, October, 2026
  Measurement Systems Security AI/ML
- *What Typos Don't Explain: An Empirical Study of Queries of Non-Existent Domain Names**. **Boladji Vinny Adjibi**, Michael Bailey, and Fabian Monrose. Submitted to ACM Web Conference, October, 2026.
  Measurement AI/ML Human Factors
- *Beyond Detection: Source Attribution of Synthetic Speech using Multilabel Classifier Chains**. **Boo Fullwood** and Fabian Monrose. Submitted to IEEE International Conference on Acoustics, Speech, and Signnal Processing, Sept, 2026.
  AI/ML Forensics Systems Security
- *Exploiting Software-level Abstractions to Support Practical Hardware Trojan Attacks**. Athanasios Moschos, Georgios Kokolakis, **Kevin Valakuzhy**, Fabian Monrose and Angelos Keromytis. ePrint: arXiv 2609.23173, September, 2026. Under revision for submission to IEEE European Symposium on Security and Privacy, Nov. 2027.
  Hardware Systems Security Side Channels

Published and Accepted Articles

- *Acoustic Microcues: Breath and Phoneme-Level Signals for Unmasking Audio Deepfakes**. **Lydia Han, Amogh Palasamudram, Boo Fullwood**, Fabian Monrose. Deepfake Forensics Workshop: Detection, Attribution, Recognition, and Adversarial Challenges in the Era of AI-Generated Media, November, 2026. Biometrics AI/ML Forensics
- *BRAID: Uncovering Malware Family Relationships in Data-Scarce Settings**. **Kevin Valakuzhy**, Miuyin Yong Wong, Omar Alrawi, Manos Antonakakis, Angelos D. Keromytis and Fabian Monrose. European Symposium on Research in Computer Security (ESORICS), September, 2026.
  Malware Systems Security Forensics
- *HeartBreaker: Practical Attacks for Recovering Cryptographic Keys Derived from Biosignals**. Evangelos Froudakis and Fabian Monrose. European Symposium on Research in Computer Security (ESORICS), September, 2026.
  Privacy Biometrics AI/ML
- *The Impact of Emerging AI Practices on the Cybersecurity Workforce*. Miuyin Wong, Alan F. Luo,

Yunzhe Zhao, **Fabian Monrose**, and Michelle Mazurek. USENIX Symposium on Usable Privacy and Security (SOUPS), August, 2026.
Human Factors Privacy Policy

- *Beyond the Stars: Multimodal Detection of Scams on GitHub**. **Tillson Galloway, Kevin Valakuzhy**, Manos Antonakakis and Fabian Monrose. USENIX Security Symposium, August, 2026.
Supply Chain Systems Security AI/ML
- *SoK: Multi-Layer Indirect Call Analysis in the Real World**. **Yufei Du**, Vasileios P. Kemerlis, Michalis Polychronakis and Fabian Monrose. USENIX WOOT Conference on Offensive Technologies, August, 2026.
Program Analysis Systems Security Software Systems
- *The Cost of Silence: Quantifying the Prejudice of Non-Participation in Domain Name Disputes**. **Boladji Vinny Adjibi**, Kathryn Kleiman, Michael Bailey, and Fabian Monrose. Int. Conference on Artificial Intelligence and Law (ICAIL), June, 2026.
Policy Measurement AI/ML
- *LUMEN: A Systems Approach to LLM-Guided Activation of Hidden Behaviors in Malware**. **Kevin Valakuzhy**, Miuyin Yong Wong, Doug Blough, Mustaque Ahamad and Fabian Monrose. International Conference on Dependable Systems and Networks (DSN), June, 2026.
Systems Security Malware Forensics
- *Mind Your [M]s, Cross Your [T]s: A large-scale Phonemic Analysis of Speech Reproduction in Modern Speech Generators**. **Boo Fullwood** and Fabian Monrose. IEEE Conference on Acoustics, Speech and Signal Processing (ICASSP), May 2026.
Acoustics Privacy AI/ML
- *Actively Understanding the Dynamics and Risks of the Threat Intelligence Ecosystem**. **Tillson Galloway, Allen Chang**, Omar Alrawi, Thanos Avgetidis, Manos Antonakakis and Fabian Monrose. Network and Distributed System Security (NDSS) Symposium, Feb 2026.
Measurement Supply Chain Systems Security
- *Repairing Trust in Domain Name Disputes Practices: Insights from a Quarter-Century's Worth of Squabbles**. **Boladji Vinny Adjibi**, Athanasios Avgeditis, Manos Antonakakis, Alberto Dainotti, Michael Bailey and Fabian Monrose. Network and Distributed System Security (NDSS) Symposium, Feb 2026.
Policy Measurement AI/ML
- *From Concealment to Exposure: Understanding the Lifecycle and Infrastructure of APT Domains**. Athanasios Avgetidis, **Boladji Vinny Adjibi**, Panagiotis Kintis, **Tillson Galloway**, Zane Ma, Omar Alrawi, Manos Antonakakis, Roberto Perdisci, Fabian Monrose and Angelos Keromytis. International Symposium on Research in Attacks, Intrusions and Defenses, October, 2025.
Measurement Systems Security AI/ML
- *The Guardian of Name Street: Studying Defensive Registrations for the Fortune 500**. **Boladji Vinny Adjibi**, Athanasios Avgetidis, Michael Bailey, Manos Antonakakis and Fabian Monrose. In Network and Distributed Systems Security (NDSS), Feb. 2025.
Measurement System Security AI/ML
- *Seeing is Believing: Interpreting Behavioral Changes in Audio Deepfake Detectors Arising from Data Augmentation**. **Boo Fullwood** and Fabian Monrose. In 18th ACM Workshop on Artificial Intelligence and Security, October, 2025.
AI/ML Forensics System Security

- *Understanding LLMs Ability to Aid Malware Analysts in Bypassing Evasion Techniques**. Miuyin Wong, **Kevin Valakuzhy**, Mustaque Ahamad, Douglas M. Blough, and Fabian Monrose. In International Conference on Multimodal Interaction (ICMI), 2024.
  Malware AI/ML Forensics
- *Contrasting Theory to Practice: Results from Interviews with Expert Malware Analysts**. Miuyin Wong, Matthew Landen, Frank Li, Fabian Monrose and Mustaque Ahamad. In USENIX Symposium on Usable Privacy and Security (SOUPS), August, 2024.
  Human Factors Malware Policy
- *CrashTalk: Automated Generation of Precise, Human Readable, Descriptions of Software Security Bugs**. Kedrian James, **Kevin Valakuzhy**, Kevin Snow and Fabian Monrose. In ACM Conference and Computer and Data Security (CODASPY), June, 2024.
  Malware Systems Security Measurement
- *Towards Practical Fabrication Stage Attacks Using Interrupt-Resilient Hardware Trojans**. Athanasios Moschos, Fabian Monrose and Angelos Keromytis. In IEEE Int. Symposium on Hardware Oriented Security and Trust (HOST), May, 2024.
  Hardware Systems Security Program Analysis
- *Improving Security Tasks Using Compiler Provenance Information Recovered At the Binary-Level**. **Yufei Du**, Omar Alwari, Kevin Snow, Manos Antonakakis and Fabian Monrose. In ACM Conference on Computer and Communications Security (CCS), November, 2023.
  Program Analysis Forensics Measurement
- *Stale TLS Certificates: Investigating Precarious Third-Party Access to Valid TLS Keys**. Zane Ma, Aaron Faulkenberry, Thomas Papastergiou, Zakir Durumeric, Michael D. Bailey, Angelos D. Keromytis, Fabian Monrose and Manos Antonakakis. In Proceedings of ACM Internet Measurement Conference (IMC), Oct, 2023.
  Measurement Networks System Security
- *More Carrot or Less Stick: Organically Improving Student Time Management With Practice Tasks and Gamified Assignments*. Mac Malone and Fabian Monrose. In Proceedings of ACM Conference on Innovation and Technology In Computer Science Education, July, 2023.
  Education Systems Security Human Factors
- *Securely Autograding Cybersecurity Exercises Using Web Accessible Jupyter Notebooks*. Mac Malone, Yicheng Wang and Fabian Monrose. In Proceedings of ACM SIGCSE Technical Symposium, March, 2023.
  Education Measurement Human Factors
- *Beyond the Gates: An Empirical Analysis of HTTP-managed password stealers and operators**. Omar Alrawi, Athanasios Avgetidis, **Kevin Valakuzhy**, Charles Lever, Paul Burbage, Angelos Keromytis, Fabian Monrose, and Manos Antonakakis. In Proceedings of the USENIX Security Symposium, August 2023.
  Forensics Privacy Measurement
- *View from Above: Exploring the Malware Ecosystem from the Upper DNS Hierarchy**. Aaron Faulkenberry, Athanasios Avgetidis, Zane Ma, Omar Alrawi, Charles Lever, Panagiotis Kintis, Fabian Monrose, Angelos Keromytis, and Manos Antonakakis. In Proceedings of the Annual Computer Security Applications Conference (ACSAC), Dec 2022.
  Measurement Malware Forensics
- *Automatic Recovery of Fine-grained Compiler Provenance Information at the Binary Level*. **Yufei Du**, Ryan Court, Kevin Z. Snow and **Fabian Monrose**. USENIX Annual Technical Conference, July, 2022.

Program Analysis Forensics Measurement

- *Disentangling Style and Content for Low Resource Video Domain Adaptation: A Case Study on Keystroke Inference Attacks*. John Lim, Jan-Michael Frahm and Fabian Monrose. In ACM Conference on Data and Application Security (CODASPY), April, 2022.
  Biometrics System Security Privacy
- *An Online Gamified Learning Platform for Teaching Cybersecurity and More*. Mac Malone, Yicheng Wang and fFabian Monrose. ACM Annual Conference on IT Education, October, 2021.
  Education Human Factors AI/ML
- *DynPTA: Combining Static and Dynamic Analysis for Practical Selective Data Protection*. Tapti Palit, Jarin Firose Moon, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2021.
  Software Security Systems Security Measurement
- *Applicable Micropatches and Where to Find Them: Finding and Applying New Security Hot Fixes to Old Software*. Mac Malone, Yicheng Wang, Kevin Snow and Fabian Monrose. IEEE International Conference on Software Testing, Verification and Validation, April, 2021.
  Systems Security Measurement AI/ML
- *The Circle Of Life: A Large-Scale Study of The IoT Malware Lifecycle**. Omar Alrawi, Charles Lever, **Kevin Valakuzhy**, Ryan Court, Kevin Snow, Fabian Monrose and Manos Antonakakis. USENIX Security Symposium, August, 2021.
  Measurement Malware Forensics
- *To Gamify or Not? On Leaderboard Effects, Student Engagement, and Learning outcomes in a Cybersecurity Intervention*. Mac Malone, Yicheng Wang, Kedrian James, Murray Anderegg, Jan Werner and Fabian Monrose. ACM SIGCSE Technical Symposium, 2021.
  Education Human Factors System Security
- *A Flexible Framework for Expediting Bug Findings by Leveraging Past (Mis-)Behavior to Discover New Bugs*. Sanjeev Das, Kedrian James, Jan Werner, Manos Antonakakis, Michalis Polychronakis and Fabian Monrose. ACM Annual Computer Security Applications Conference (ACSAC), December 2020.
  Systems Security Measurement Forensics
- *Methodologies for Quantifying (Re-)randomization Security and Timing under JIT-ROP*. Md Salman Ahmed, Ya Xiao, Kevin Z. Snow, Gang Tan, Fabian Monrose and Danfeng Yao. In Proceedings of the ACM Conference on Computer and Communications Security, October, 2020.
  Software Security Systems Security Measurement
- *Mitigating Data Leakage by Protecting Memory-resident Sensitive Data*. Tapti Palit, Fabian Monrose and Michalis Polychronakis. In Proceedings of the Annual Computer Security Application Conference, December 2019.
  Systems Security Software Security Privacy
- *The SEVerESt Of Them All: Inference Attacks Against Secure Virtual Enclaves*. Jan Werner, Joshua Mason, Manos Antonakakis, Michalis Polychronakis and Fabian Monrose. In Proceedings of the ACM Asia Conference on Computer and Communication Security, July 2019.
  Systems Security Forensics AI/ML
- *SoK: The Challenges, Pitfalls, and Perils of Using Hardware Performance Counters for Security*. S.

Das, J. Werner, M. Antonakakis, M. Polychronakis and F. Monrose. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2019.
Privacy Systems Security Measurement

- *SoK: Security Evaluation of Home-Based IoT Deployments**. O. Alrawi, C. Lever, M. Antonakakis and F. Monrose. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2019.
Privacy Systems Security Measurement
- *Security Concerns in Asynchronous Web Server Architectures: When Performance Optimizations Empower Data-oriented Attacks*. M. Morton, J. Werner, K. Z. Snow, M. Antonakakis, M. Polychronakis and F. Monrose. In Proceedings of the IEEE European Symposium on Security and Privacy, April, 2018.
Software Security Systems Security Measurement
- *Practical Attacks Against Graph-based Clustering**. Y. Chen, Y. Nadji, A. Kountouras, F. Monrose, R. Perdisci, M. Antonakakis, N. Vasiloglou. In Proceedings of the ACM Conference on Computer and Communications Security, October, 2017.
AI/ML Systems Security Measurement
- *Keeping Zombie Gadgets at Bay by Re-randomizing after Disclosure*. Micah Morton, Forrest Li, Kevin Snow, Michalis Polychronakis and Fabian Monrose. In Proceedings of Engineering Secure Systems and Software conference (ESSoS), July, 2017.
Systems Security Measurement Software Security
- *Revisiting Browser Security in the Modern Era: New Data-only Attacks and Defenses*. Roman Rogowski, Micah Morton, Forrest Li, Kevin Z. Snow, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE European Symposium on Security and Privacy, March 2017.
Privacy Systems Security Software Security
- *Virtual U: Defeating Face Liveness Detection by Building Virtual Models From Your Public Photos*. Yi Xu, True Price, Jan-Michael Frahm and Fabian Monrose. In Proceedings of the USENIX Security Symposium, August 2016.
Systems Security Privacy AI/ML
- *Return to the Zombie Gadgets: Undermining Destructive Code Reads via Code-Inference Attacks*. Kevin Z. Snow, Roman Rogowski, Jan Werner, Hyungjoon Koo, Fabian Monrose and Michalis Polychronakis. In Proceedings of the IEEE Symposium on Security and Privacy, May, 2016.
Software Security Systems Security Measurement
- *No-Execute-After-Read: Preventing Code Disclosures in Commodity Software*. Jan Werner, George Baltas, Rob Dallara, Nathan Otterness, Kevin Snow, Fabian Monrose and Michalis Polychronakis. In Proceedings of the ACM Asia Conference on Computer and Communication Security, May 2016.
Systems Security Measurement Privacy
- *Detecting Malicious Exploit Kits using Tree-based Similarity Searches*. Teryl Taylor, Xin Hu, Ting Wang, Jiyong Jang, Marc Stoeckin, Fabian Monrose and Reiner Sailer. In Proceedings of the ACM Conference on Data and Application Security and Privacy, pages 255-266, March, 2016.
Malware Privacy Measurement
- *Cache, Trigger, Impersonate: Enabling Context-Sensitive Honeyclient Analysis On-the-Wire*. Teryl Taylor, Kevin Snow, Nathan Otterness and Fabian Monrose. In Proceedings of the 23nd ISOC Network and Distributed Systems Security Symposium (NDSS), Feb., 2016.

Systems Security Measurement Software Security

- *Isomeron: Code Randomization Resilient to (Just-in-Time) Return-Oriented Programming*. Luca Davi, Christopher Liebchen, Ahmad-Reza Sadeghi, Kevin Z. Snow, and Fabian Monrose. In Proceedings of the Network and Distributed Systems Security Symposium (NDSS), Feb., 2015.
  Systems Security Measurement Software Security
- *Watching the Watchers: Inferring TV Content from Outdoor Light Effusions*. Yi Xu, Jan-Michael Frahm and Fabian Monrose. In Proceedings of the 21st ACM Conference on Computer and Communications Security (CCS), November, 2014.
  Systems Security Privacy AI/ML
- *Emergent Faithfulness to Morphological and Semantic Heads in Lexical Blends*. Katherine Shaw, Elliott Moreton, Andrew White and Fabian Monrose. In Proceedings of the Annual Meeting on Phonology, February, 2014.
  Phonology Human Factors Linguistics
- *Security and Usability Challenges of Moving-Object CAPTCHAs: Decoding Codewords in Motion*. Yi Xu, Gerardo Reynaga, Sonia Chiasson, Jan-Michael Frahm and Fabian Monrose. IEEE Transactions on Dependable and Secure Computing, February, 2014.
  Human Factors Biometrics Privacy
- *On the Privacy Risks of Virtual Keyboards: Automatic Reconstruction of Typed Input from Compromising Reflections*. Rahul Raguram, Andrew White, Yi Xu, Jan-Michel Frahm, Pierre Georgel, and Fabian Monrose. IEEE Transactions on Dependable and Secure Computing, Volume 10, Issue 3, pages 154-167, May-June, 2013.
  Privacy Systems Security Human Factors
- *Seeing Double: Reconstructing Obscured Typed Input from Repeated Compromising Reflections*. Yi Xu, Jared Heinly, Andrew M. White, Fabian Monrose and Jan-Michael Frahm. In Proceedings of the ACM Conference on Computer and Communications Security (CCS), November, 2013.
  Privacy Systems Security Measurement
- *Check my profile: Leverage static analysis for fast and accurate detection of ROP gadgets*. Blaine Stancill, Kevin Snow, Nathan Otterness, Fabian Monrose, Lucas Davi, and Ahmad- Reza Sadeghi. In Proceedings of the 16th International Symposium on Research in Attacks, Intrusions, and Defenses, October, 2013.
  Systems Security Measurement Software Security
- *Crossing the Threshold: Detecting Network Malfeasance via Sequential Hypothesis Testing*. Srinivas Krishnan, Teryl Taylor, Fabian Monrose and John McHugh. In Proceedings of the 42nd Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Performance and Dependability Symposium, June, 2013.
  Systems Security Measurement AI/ML
- *Just-In-Time Code Reuse: On the Effectiveness of Fine-Grained Address Space Layout Randomization*. Kevin Snow, Lucas Davi, Alexandra Dmitrienko, Christopher Liebchen, Fabian Monrose and Ahmad-Reza Sadeghi. In Proceedings of 34th IEEE Symposium on Security and Privacy, May, 2013. (**Best Student Paper Award**).
  Privacy Systems Security AI/ML
- *Clear and Present Data: Opaque Traffic and its Security Implications for the Future*. Andrew White, Srinivas Krishnan, Michael Bailey, Fabian Monrose and Phil Porras. In Proceedings of the 20th ISOC Network and Distributed Systems Security Symposium, Feb., 2013.

Systems Security Measurement AI/ML

- *Security and Usability Challenges of Moving-Object CAPTCHAs: Decoding Codewords in Motion*. Yi Xu, Gerardo Reynaga, Sonia Chiasson, Jan-Michael Frahm, Fabian Monrose and Paul van Oorschot. In Proceedings of the 21th USENIX Security Symposium, August, 2012.
  Human Factors Biometrics Privacy
- *Toward Efficient Querying of Compressed Network Payloads*. Teryl Taylor, Scott E. Coull, Fabian Monrose and John McHugh. In USENIX Annual Technical Conference, June, 2012.
  Measurement Systems Security AI/ML
- *Trail of Bytes: New Techniques for Supporting Data Provenance and Limiting Privacy Breaches*. Srinivas Krishnan, Kevin Snow, and Fabian Monrose. IEEE Transactions on Information Forensics and Security, pages 1876-1889, Issue 6, Volume 7, 2012.
  Forensics Privacy Human Factors
- *Understanding Domain Registration Abuses (Invited Paper)*. Scott Coull, Andrew White, Ting-Fang Yen, Fabian Monrose and Michael K. Reiter. Special Issue of Computers & Security (COSE), pages 806-815, Volume 31, Number 7, October, 2012.
  Measurement Systems Security AI/ML
- *iSpy: Automatic Reconstruction of Typed Input from Compromising Reflections*. Rahul Raguram, Andrew White, Dibyendusekhar Goswami, Fabian Monrose and Jan-Michael Frahm. In Proceedings of the 18th ACM Conference on Computer and Communications Security (CCS), November, 2011.
  Privacy Systems Security Measurement
- *ShellOS: Enabling Fast Detection and Forensic Analysis of Code Injection Attacks*. Kevin Snow, Srinivas Krishnan, Fabian Monrose and Niels Provos. In Proceedings of the 20th USENIX Security Symposium, August, 2011.
  Forensics Systems Security Software Systems
- *An Empirical Study of the Performance, Security and Privacy Implications of Domain Name Prefetching*. Srinivas Krishnan and Fabian Monrose. In Proceedings of the 41st Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Dependable Computing and Communications Symposium, June, 2011.
  Measurement Systems Security Software Security
- *Amplifying Limited Expert Input to Sanitize Large Network Traces*. Xin Huang, Fabian Monrose, and Michael K. Reiter. In Proceedings of the 41st Annual IEEE/IFIP International Conferences on Dependable Systems and Networks; Performance and Dependability Symposium, June, 2011.
  Systems Security Measurement AI/ML
- *Phonotactic Reconstruction of Encrypted VoIP Conversations*. Andrew White, Kevin Snow, Austin Matthews and Fabian Monrose. In Proceedings of 32nd IEEE Symposium on Security and Privacy, May, 2011. (**Best Paper Award**).
  Privacy System Security Measurement
- *Towards Optimized Probe Scheduling for Active Measurement Studies*. Daniel Kumar, Fabian Monrose and Michael K. Reiter. In Proceedings of the 6th International Conference on Internet Monitoring and Protection (ICIMP), March, 2011 (**Best Paper Award**).
  Measurement Privacy Networks
- *On Measuring the Similarity of Network Hosts: Pitfalls, New Metrics, and Empirical Analyses*. Scott Coull, Micheal Bailey and Fabian Monrose. In Proceedings of the 18th Annual Network

and Distributed Systems Security Symposium, February, 2011.
Measurement Privacy Networks

Provisional Patents, Applications, and Invention Disclosures

- **Vinny Adjibi**, Michael D. Bailey and Fabian Monrose. *AI-Based System for Improved Brand Protection*, U.S. Provisional Patent Filing No. 64/167,370, October 1, 2026.
- **Kevin Valakuzhy**, Miuyin Yong Wong, Omar Alrawi, Manos Antonakakis, Angelos D. Keromytis and Fabian Monrose. *Systems and Methods for Malware Similarity via Datastore of Assembly-level Malicious Behaviors Extracted from Memory*, US Provisional Patent Filing No. 19/380,650, November 5, 2025.
- **Kevin Valakuzhy, Miuyin Yong Wong**, Fabian Monrose, Douglas Blough and Mustaque Ahmad. *LLM-Enabled System for Locating and Bypassing Evasive Malware Tactics*, U.S. Provisional Patent Filing No. 63/898,908, October 14, 2025.

Patents Awarded

- Jan Werner, Rob Dallara, George Baltas, Michalis Polychronakis, Kevin Snow and Fabian Monrose. *Methods, Systems, and Computer Readable Media for Preventing Code Reuse Attacks.* U.S. Patent No. 10,628,589, April, 2020.
- Xin Hu, Jiyong Jang, Fabian Monrose, Marc Philippe Stoecklin, Teryl Taylor and Ting Wang. *Detecting Web Exploit Kits by Tree-based Structural Similarity Search*. U.S. Patent No. 10,560,471, Feb, 2020.
- T. Taylor, K. Snow, N. Otterness, and F. Monrose. *Methods, systems, and computer readable media for detecting malicious network traffic*, U.S. Patent No. 9,992,217, June 2018.
- Srinivas Krishnan, Fabian Monrose, Michael Bailey, Phillip Porras, and Andrew White. *Methods, Systems, and Computer Readable Media for Rapid Filtering of Opaque Data Traffic*, U.S. Patent No. 9,973,473, May 2018.
- Srinivas Krishnan, Teryl Taylor, Fabian Monrose and John McHugh. *Methods, Systems, and Computer Readable Media for Detecting Compromised Computing Hosts*. U.S. Patent No. 9,934,379, April 2018.
- Srinivas Krishnan, Kevin Snow, and Fabian Monrose. *Methods, Systems, and Computer Readable Media for Efficient Computer Forensic Analysis and Data Access Control*. U.S. Patent No. 9,721,089, August 2017.
- Kevin Z. Snow, Fabian Monrose and Srinivas Krishnan. *Methods, Systems, and Computer Readable Media for Detecting Injected Machine Code*. U.S. Patent 9,305,165, April 5, 2016.
- Fabian Monrose, Teryl Taylor, Srinivas Krishnan and John McHugh. United States Patent Application Serial No. 14/773,660 for *Methods, Systems, and Computer Readable Media for Detecting a Compromised Computing Host*, March, 2015.

Grants and Contracts (*last 10 years only*)

PI, DARPA (Jim Simpson (*lead, Cynnovative*)), STTR Phase II: Learning an Unknown Compiler for Instant Development (LUCID); share $540,000. April 2026 – May 2028.

Co-PI, Verisign Inc. (with M. Bailey), Securing DNS; share $128,000. Feb 2026 – Feb 2027.

Co-PI, ARPA-H (Brendon Saltaformaggios (*lead*)), Hospital-Integrated Vulnerability Identification and Proactive Remediation (H-VIPER); share $402,000. October 2025 – May

2027.

Co-PI, NAVY/National Security Technology Accelerator (K. Bittick at GTRI (*lead*)), Firmware Bill of Materials Extractor; share $684,000. August 2024 – July 2027.

Senior Personnel, DARPA (M. Antonakakis at GaTech (*lead*)), A Novel Framework to Assess, Statistically Model and Evade Cyber Attack Infrastructure; share $1,266,714, October 2022 — April 2027 (NCE).

Co-PI (with Jan-Michel Frahm), CIA, Cyber Identity and Behavioral Analytics Research Consortium (CIBAR); share $538,000, July 2017 — Dec, 2021.

PI, DARPA (M. Antonakakis at GaTech (*lead*)), Attributing Cyber Actors through Tensor Decomposition and Novel Data Acquisition; share $2,019,181, Nov. 2016 — August 2021.

PI, NIST, Knowledge is Power: Assessing the Security and Privacy of Consumer IoT Architectures and Systems for $183,063.00, July 2017 — July, 2018.

PI, IEEE, Advancing Cybersecurity Education Through Active, Challenge-Based, Learning Exercises for $119,999.44, Jan. 2017 — August, 2018.

PI, Department of Defense University Research Instrumentation Program (DURIP), Next Generation Defenses against Web-based Exploits for $116,500.00, April, 2016 — May 2017.

PI, NSF *Secure and Trustworthy Computing*, TWC: TTP Option: Small: Scalable Techniques for Better Situational Awareness: Algorithmic Frameworks and Large Scale Empirical Analyses for $667,900.00, Sept. 2014—August 2017.

PI Department of Homeland Security *Transition to Practice Program*, Supplement to NSF SDCI Sec: New Software Platforms for Supporting Network-wide Detection of Code Injection Attacks for $350,000.00, Sept. 2014—August 2015.

[V]. Education

- **Courses Taught**
  - *Advanced Computer Security, ECE 8813*, Fall 2025, 8 students.
  - *Advanced Computer Security, ECE 8813*, Fall 2024, 14 students.
  - *Advanced Computer Security, ECE 8813*, Fall 2023, 15 students.

- **Individual Student Guidance**
  - **In Progress Ph.D. Students**
    - Lydia Han, Ph.D., *TBD*, Spring 2025 – Fall 2028
    - James Fullwood, Ph.D, *Multilabel Classification of Synthetic Audio*, Fall 2023 – Fall 2028
    - Vinny Adjibi, Ph,D, *Robust Brand Protection in the Domain Name Space*, Spring 2023 – 2028 (*Palo Alto Networks*)
    - Yufei Du, Ph.D, *Leveraging Compile-time Metadata to Detect Compiler-Induced Security Issues and Selectively Invoke Security Protections at Deploy Time*, Spring 2023 – Spring 2027
    - Kevin Valakuzhy, Ph,D., *Robust Approaches for Reducing Analyst Workloads with Agentic Systems*, Fall 2022 – Fall 2026
    - Tillson Galloway, *Demystifying Operational Gaps in the Threat Intelligence Ecosystem*, Ph.D., expected graduation: Fall 2024 – Fall 2026 (*Meta*)

– **Graduated Ph.D. Students (external to Georgia Tech)**
  * Kedrian James, Spring 2024, Ph.D., UNC Chapel Hill (*Highpoint University*)
  * Mac Malone, Fall 2023, Ph.D., UNC Chapel Hill (*Lean*)

– **M.S. Students**
  * Amogh Palasamudram, MSe, Summer 2024 – Spring 2026 (*Google*)
  * Lindsey Lubin, *Leveraging Large Language Models to Support Individualized Learning*, Fall 2024 – Spring, 2025 (*Sandia National Labs*)

– **Undergraduate Students**
  * Allen Chang, Summer 2023 – Fall, 2025 (*Palo Alto Networks*)

– **Service on Thesis or Dissertation Committees (* = Georgia Tech)**
  * Athanasios Moschos, *A Systems Perspective on Hardware Trojans Inside Processors**, Ph.D., Georgia Institute of Technology, Advisor: Angelos Keromytis, 2026 (expected)
  * Arman Allahverdi, *Design, Analysis, and Implementation of a Scalable Hardware-Oriented Cryptographic Primitive**, Ph.D., Georgia Institute of Technology, Advisor: Vincent J. Mooney, 2026 (expected)
  * Athanasios Avgetidis, *Towards Understanding the Lifecycle of Malicious Network Infrastructure**, Ph.D., Georgia Institute of Technology, Advisor: Manos Antonakakis, 2025
  * Aaron Faulkenberry, *Internet-Scale Discovery and Measurement of DNS Monitoring for Cyber Threats**, Ph.D., Advisor: Manos Antonakakis Georgia Institute of Technology, 2025
  * Kedrian James, *Improving Software Security Triage and Remediation*, Ph.D., UNC Chapel Hill, May, Advisor: Fabian Monrose, 2024.
  * Miuyin M. Wong, *Narrowing the Gap between Research and Practice through the Understanding of Malware Analysis Workflows**, Ph.D., Georgia Institute of Technology, Advisor: Mustaque Ahmad, 2024
  * Kevin Hutto, *Remote Sensor Security Through Encoded Computation and Cryptographic Signatures**, Ph.D., Advisor: Vincent J. Mooney, Georgia Institute of Technology, May, 2024.
  * Mac Malone, *Improving Student Engagement and Learning Outcomes with a Gamified Learning Platform*, Ph.D., UNC Chapel Hill, Advisor: Fabian Monrose, May, 2023.
  * Omar Alrawi, *Security and Threat Evaluation of Smart Home Technology**, Georgia Institute of Technology, Advisor: Manos Antonakakis, July, 2022.
  * Salman Ahmed, *Quantitative Metrics and Measurement Methodologies for System Security Assurance*, Virginia Tech, Advisor: Daphne Yao, Dec, 2021.
  * Tapti Palit, *Sensitive Data Encryption: A Scalable Defense against Sensitive Data Leakage*, Stony Brook University, Advisor: Michalis Polychronakis, October, 2021.
  * Calvin Deutschbein, *Mining Secure Behavior of Hardware Designs*, UNC Chapel Hill, Advisor: Cynthia Sturton, July, 2021.

– **Mentorship of Research Scientists**
  * Omar Alwari, 2023 – present.
  * Panagiotis Kintis, 2024 – August 2026.

SERVICE
(*last 10 years***)**

- **Professional Contributions**
  - *Distinguished Paper Awards Committee*, 33rd ACM CCS, 2026
  - *TPC*, 32nd – 33rd ACM Conference on Computer and Communications Security, 2025-2026
  - *TPC*, 47th IEEE Symposium on Security and Privacy, 2026
  - *Test of Time Awards Committee*, USENIX Security Symposium, 2022-2024
  - *TPC*, 14th ACM Conference on Data and Application Security and Privacy, 2024
  - *TPC*, 13th ACM Conference on Data and Application Security and Privacy, 2023
  - *TPC*, 29th Annual Computer Security Applications Conference, 2021
  - *TPC*, 40th – 41st IEEE Symposium on Security and Privacy, 2019-2020
  - *TPC*, 20th Symposium on Research in Attacks and Defenses, 2018
  - *TPC*, CVPR Workshop on Computer Vision and Computer Security, 2017
  - *TPC*, 18th Symposium on Research in Attacks and Defenses, (**Chair**), 2016
- **Institute Contributions**
  - Co-director, ECE Center for Cyber Operations Enquiry and Unconventional Sensing (CEOUS), 2024 – present
  - ECE PhD *Admissions committee* (Area Chair for Software & Systems TIG), 2023–2027
  - *Faculty Recruiting Committee*, School of Cybersecurity & Privacy, 2023–2025
  - *COE Cybersecurity Fellows Selection Committee Chair*, 2023